Tumour-Infiltrating CHIP in CRC, HNSCC and PDAC: Shared Genomic Patterns and Immune Microenvironment Signal

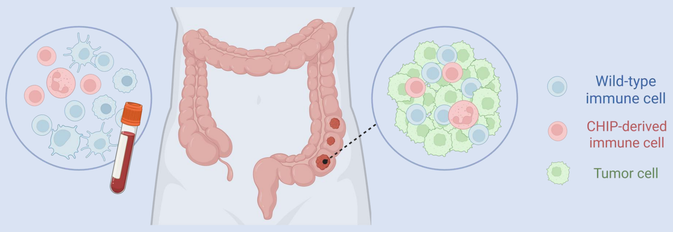
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Introduction

Clonal haematopoiesis of indeterminate potential (**CHIP**) is increasingly recognised as a phenomenon linked to ageing, inflammation, and the emergence of confounding variants in liquid biopsy analyses.

Tumour-Infiltrating CHIP (**TI-CH**), are CHIP-associated mutations originating from haematopoietic clones but detected within immune cells infiltrating the tumour microenvironment.



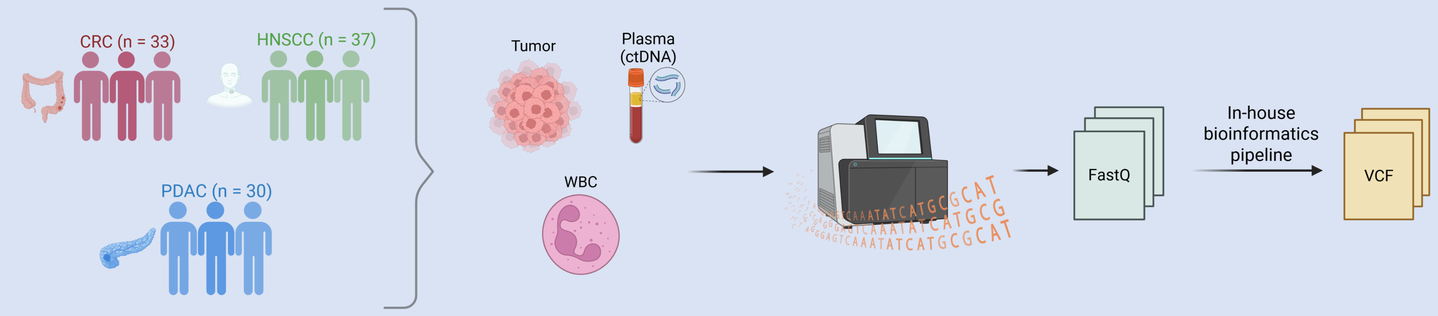
TI-CH variants may influence tumour-immune interactions, stromal remodelling and patient outcomes, but their functional significance remains poorly understood.

Objectives

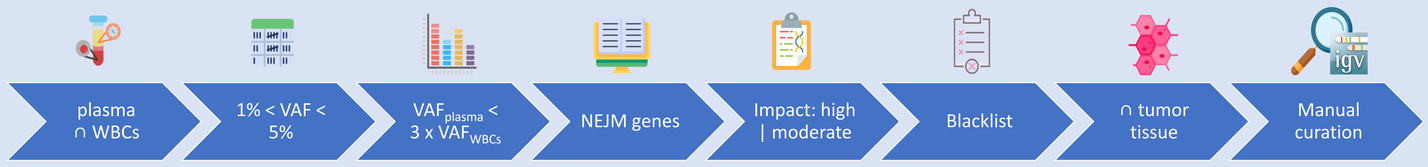
1. To characterise the prevalence, mutational spectrum, and clinical relevance of TI-CH across three cohorts using paired plasma-WBC samples and matched tumour tissue.
2. To determine whether TI-CH presence correlates with specific TME immune compositions in CRC.

Methodology

Three independent cohorts with **paired plasma-WBC** sequencing were analysed: **CRC** (n=33), **HNSCC** (n=37) and **PDAC** (n=30). A subset also had **matched tumour tissue** available (CRC n=29; HNSCC n=31; PDAC n=13).



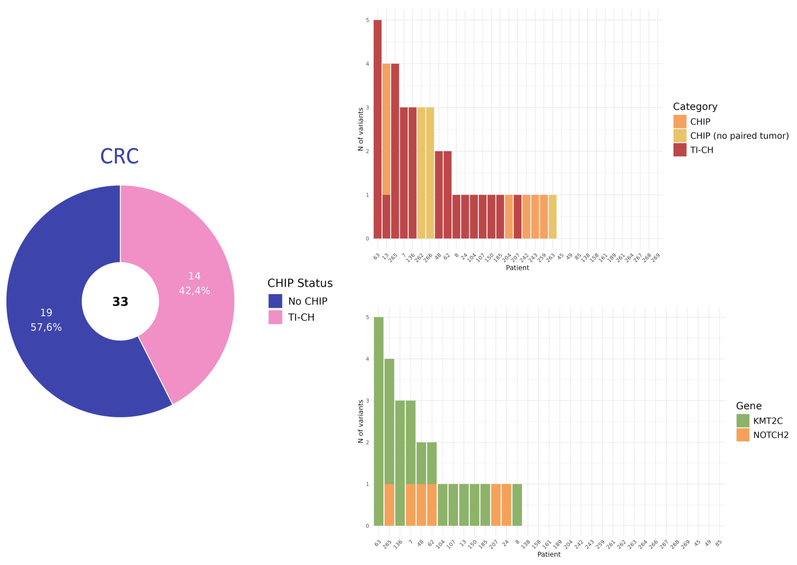
Plasma **variants** with a **VAF of 1-5%** were screened for CHIP-associated mutations and classified as CHIP if the same variant was present in both plasma and matched WBC, using the **curated gene list** from Pitch et al. (2025, NEJM). Variants were traced in matched tumour samples and **manually reviewed** using IGV to confirm their localisation within the tumour-infiltrating immune compartment rather than in the malignant clone, thereby defining TI-CH.

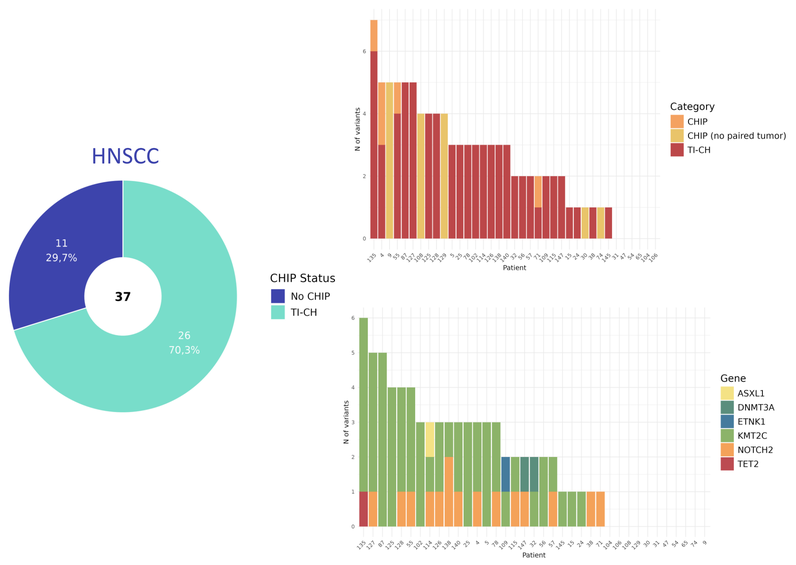


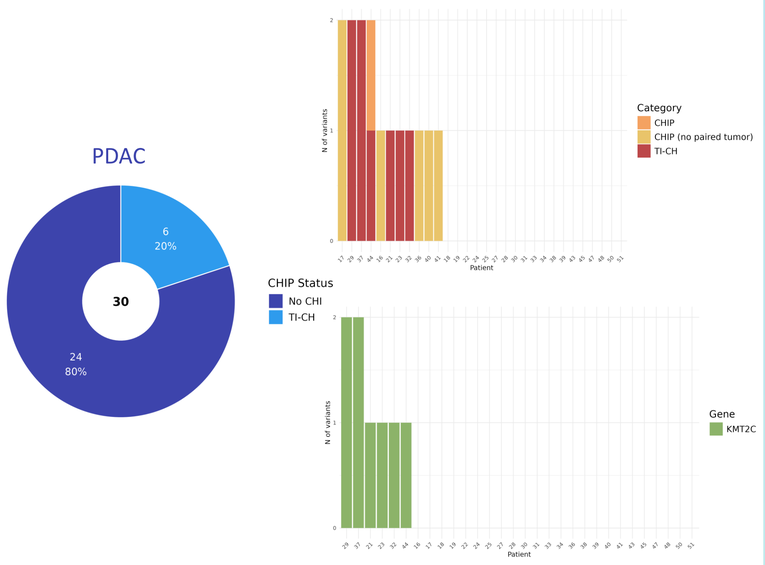
Prevalence, gene distribution and prognostic impact were assessed using **multivariable Cox** analysis. To investigate the biological significance of these findings and to evaluate whether TI-CH-positive tumours exhibited distinct immune infiltration patterns, **cellular deconvolution** was performed in CRC RNA-Seq using xCell2.

Results

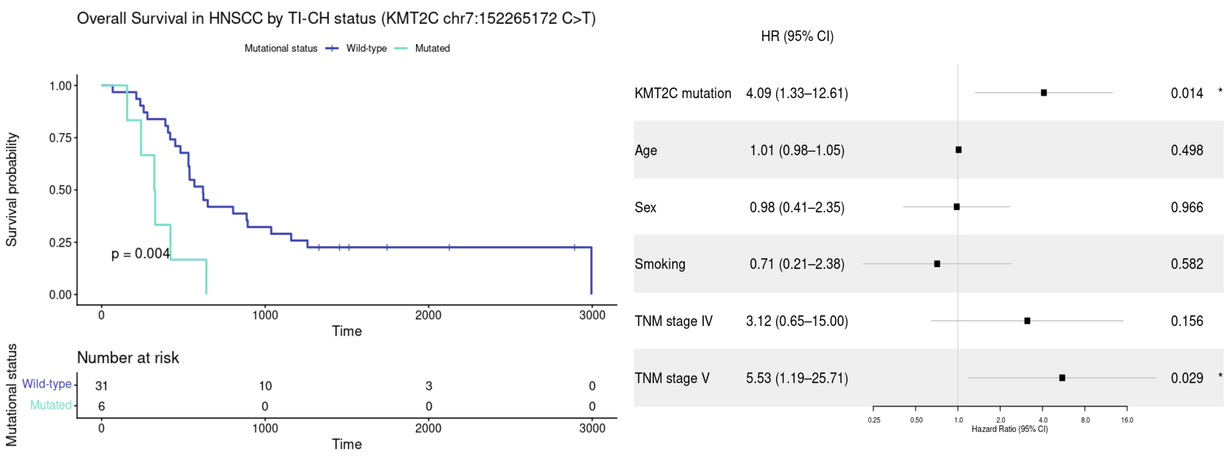
Across all tumour types, TI-CH displayed a **highly conserved molecular architecture** dominated by alterations in the **epigenetic regulator KMT2C**. Another commonly found mutated gen was NOTCH2. Contributions from classical CHIP genes, such as TET2, ASXL1, ETNK1 and DNMT3A, were infrequent.







A single hotspot mutation, **KMT2C chr7:152265172 C>T**, was detected in all three cohorts, yet only in HNSCC did it show an **association with adverse outcomes**.



TI-CH-positive CRC tumours displayed an increase in **γδ T-cell** infiltration compared with TI-CH-negative tumours. A similar increase was observed in **CD19⁺CD20⁺ B-cell** populations.

Cell type	p value	Mean E. Score TI-CH⁺	Mean E. Score no TI-CH
Enteric glial cell	0.019	2.000e-05	0,000e-0
CD19⁺ CD20⁺ B-cell	0.026	8,875e-04	7,500e-06
Stalk-like endotelial cell	0.030	8.315e-04	1.173e-04
Stromal – 1	0.030	9.975e-04	1,100e-04
Stromal – 2	0.030	5,527e-04	1,577e-04
Stem-like/Transit-amplifying epihtelial cell	0.030	3,025e-02	1,551e-03
γδ T-cell	0.039	3,5899e-23	3,739e-27
Goblet cell of colon	0.044	1,136e-22	2,409e-22

Conclusions

TI-CH shows a consistent genomic pattern across CRC, HNSCC and PDAC, dominated by KMT2C alterations and a recurrent C>T hotspot present in all three cohorts. This hotspot was associated with poorer survival in HNSCC. In CRC, TI-CH correlated with immune infiltration enriched in γδ T-cells and CD19⁺CD20⁺ B-cells, lymphocyte subsets characteristic of inflamed and remodelling tumour microenvironments. These observations suggest that TI-CH may reflect both systemic clonal dynamics and local immune reshaping. While indicative, these findings require validation in larger and more diverse cohorts to establish their broader significance.